

Heat and Mass Transfer

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Syllabus

ME 1604 : HEAT AND MASS TRANSFER

Module I

Introduction to heat transfer – basic modes of heat transfer – conduction heat transfer – Fourier law of heat conduction – temperature dependence of thermal conductivity- general heat conduction equation in cartesian, cylindrical and spherical coordinates – boundary conditions – one-dimensional steady state conduction- critical insulation thickness -one-dimensional steady state conduction with heat generation -extended surface – two dimensional steady state heat conduction – conduction shape factor – unsteady state heat conduction in one dimension – lumped heat capacity system – semi-infinite solids with sudden change in surface temperature – Introduction to numerical methods in conduction.

Module II

Convective heat transfer – Newton's law of cooling – Prandtl number – laminar forced convection heat transfer from flat plates – fully developed laminar flow in pipes – turbulent forced convection – Reynolds' analogy – natural convection – natural convection heat transfer from vertical plates and horizontal tubes – condensation and boiling – film and drop wise condensation – film boiling and pool boiling – introduction to multiphase flow and heat transfer. Diffusion and convective mass transfer-Ficks law of diffusion.

Module III

Radiative transfer – electromagnetic radiation spectrum – thermal radiation – radiation properties- black body, gray body – monochromatic and total emissive power – Planck's law – Stefan-Boltzman law – Wien's displacement law – Kirchhoffs identity – shape factor- reciprocity relation – heat exchange between non black bodies; surface and shape resistances- electrical network analogy- heat transfer between parallel surfaces – radiation shields.

Module IV

Heat Exchangers: Type of heat exchangers- overall heat transfer coefficient -fouling factors -Logarithmic mean temperature difference (LMTD)- derivation of LMTD for parallel flow and counter flow heat exchangers-LMTD correction factor- effectiveness, NTU method of heat exchanger analysis- effectiveness derivation for parallel flow and counter flow heat exchangers. Design of parallel flow-counterflow-shell and tube multipass heat exchangers-condensers.

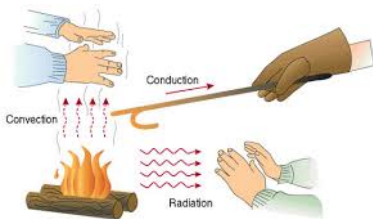
References:

- 1) Cengel, Heat Transfer Tata McGraw Hill
- 2) Holman J.P., "Heat Transfer", McGraw Hill International Students Edition
- 3) Incropera F.P. & De Witt D.P., "Fundamentals of Heat and Mass Transfer", John Wiley
- 4) Kreith F., "Heat Transfer", International Text Book Company
- 5) Gebhart B., "Heat Transfer", McGraw Hill
- 6) R.K. Rajput, "Heat and Mass Transfer", S Chand.
- 7) Venkanna, Fundamentals of HMT, PHI

Introduction

- 1 Thermodynamics $\implies$ Relation between heat and other forms of energy
- 2 Heat Transfer $\implies$ Rate of heat transfer in a system
- 3 Heat transfer rate form of Energy transfer (W)
 - 1 Conduction
 - 2 Convection
 - 3 Radiation
- 4 Potential $\implies$ Temperature difference (K)

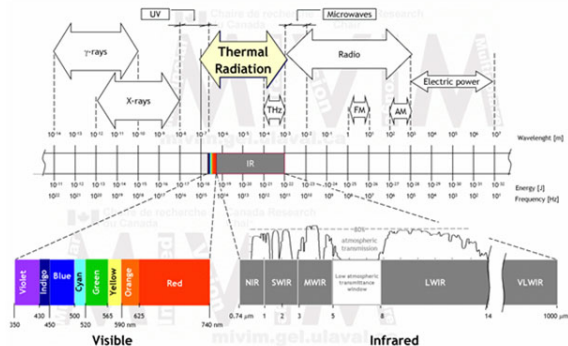
Radiation heat transfer



Radiation

- 1 Thermal Radiation- Emitted by a body as a result of its Temperature
- 2 Propagate at the speed of light

Radiation spectrum



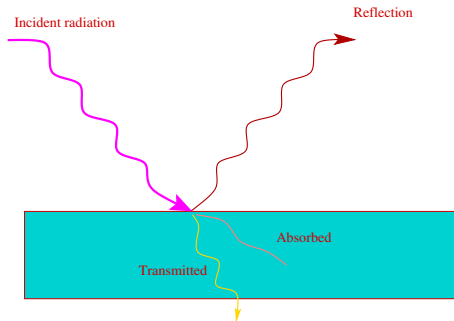
Radiation

- 1 Thermal Radiation- $0.1\mu\text{m} - 100\mu\text{m}$
- 2 Visible spectrum $0.35\mu\text{m} - 0.75\mu\text{m}$

Radiation properties

Radiation

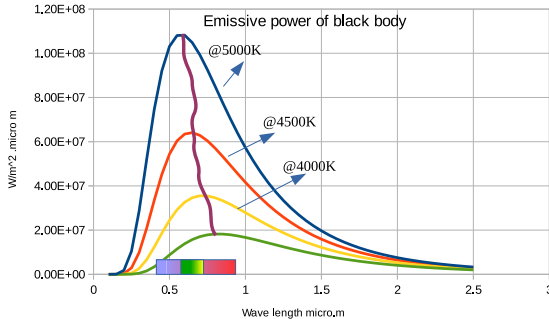
- 1 Reflectivity ρ fraction reflected
- 2 Absorptivity α fraction absorbed
- 3 Transmissivity τ fraction transmitted
- 4 $\rho + \alpha + \tau = 1$
- 5 Most solid bodies do not transmit
- 6 $\rho + \alpha = 1$



Stefan Boltzmann Law

- 1 Ideal radiator
- 2 Radiation intensity - spectral blackbody radiation intensity - Radiation energy emitted by a black body at Temp T , and λ through unit area per unit solid angle $I_{b\lambda}(T)$
- 3
$$I_{b\lambda}(T) = \frac{2\pi hc^2 \lambda^{-5}}{e^{hc/\lambda kT} - 1}$$
- 4 Spectral Emissive power $E_{b\lambda}$ - Energy radiated per unit time per unit area in all direction
- 5
$$E_{b\lambda}(T) = \frac{c_1 \lambda^{-5}}{e^{c_2/\lambda kT} - 1}$$
- 6 Where $c_1 = 3.743 \times 10^4 W \cdot \mu m^4 / m^2$, $c_2 = 1.4387 \times 10^4 \mu m K$,
- 7 Emissive power of black body $E_b(T) = \int_{\lambda=0}^{\infty} \frac{c_1 \lambda^{-5}}{e^{c_2/\lambda kT} - 1} d\lambda$
- 8 Emissive power of black body $E_b(T) = \sigma T^4$
- 9 Where $\sigma = 5.67 \times 10^{-8} W / m^2 K^4$

Wien's displacement law

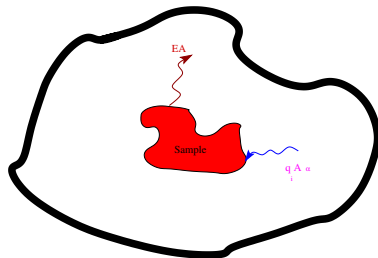


- 1 Emitted radiation increases with temperature
- 2 Peak tends to shift towards smaller wavelengths
- 3 Wien's displacement law $\lambda T_{max} = 2897.6 \mu m.K$

Kirchhoff's law

Kirchhoff's law

- 1 Emissive power of sample - E , surface area A
- 2 Energy absorbed by sample = $q_i A \alpha$
- 3 Energy emitted by sample EA
- 4 At equilibrium $EA = q_i A \alpha$
- 5 Sample is black $E_b A = q_i A 1$
- 6 Dividing $\frac{E}{E_b} = \alpha$
- 7 Ratio $\frac{E}{E_b} = \epsilon$
- 8 $\frac{E}{E_b} = \alpha = \epsilon$



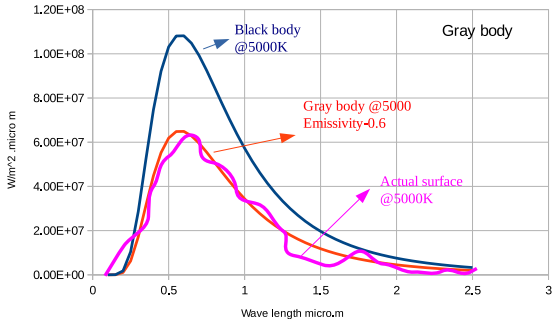
Emissivity

Total emissivities of selected materials

Material	Temp (°C)	Emissivity ϵ
Aluminum, commercial sheet	100	0.09
Copper, polished	100	0.052
Iron, dark-gray surface	100	0.31
Glass, smooth	22	0.94
Snow-white enamel varnish	23	0.906
Black shiny lacquer	24	0.875
Roofing paper	21	0.91
Porcelain, glazed	22	0.92
Red brick	23	0.93
Al-paints	100	0.27-0.67

- 1 White paint - Infra red region high emissivity
- 2 Selective surfaces -solar heater application

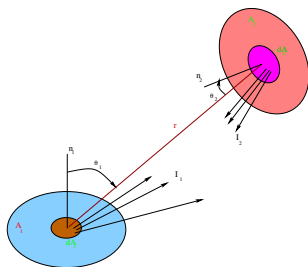
Gray body



- 1 Emissivity, absorptivity uniform over the entire wavelength
- 2 Actual body properties differ
- 3 Emissivity $\epsilon = \frac{\int_0^\infty \epsilon_\lambda E_{b\lambda}(T) d\lambda}{E_b(T)}$

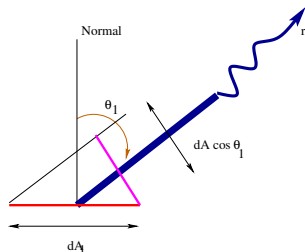
View Factor

- 1 F_{12} Fraction of the radiation energy leaving surface 1 that strike other surface 2 directly
- 2 Energy leaving surface 1 strikes 2 = $E_{b1} A_1 F_{12}$
- 3 Energy leaving surface 2 strikes 1 = $E_{b2} A_2 F_{21}$
- 4 Net energy exchange
 $Q_{1-2} = E_{b1} A_1 F_{12} - E_{b2} A_2 F_{21}$
- 5 If both surfaces are at same temperature
 $Q_{1-2} = 0 \implies E_{b1} = E_{b2}$
- 6 so that $A_1 F_{12} = A_2 F_{21}$
- 7 In general Reciprocity relation $A_i F_{ij} = A_j F_{ji}$



View Factor

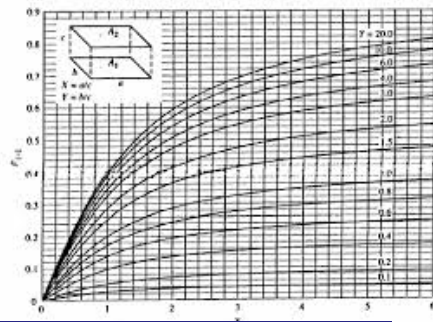
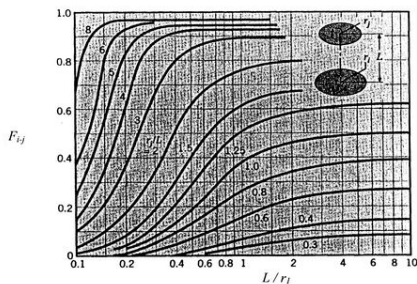
- 1 Energy leaving dA_1 in the direction $\theta_1 = I_b dA_1 \cos \theta_1$
- 2 Radiation arriving at area dA_2 at a distance r from $A_1 = I_b dA_1 \cos \theta_1 / r^2$
- 3 Black body radiation $E_b = \pi I_b$
- 4 Projected area of dA_2 receiving the radiation $= \cos \theta_2 dA_2$
- 5 Energy leaving dA_1 arriving at $dA_2, dq_{1-2} = I_{b1} \cos \theta_1 \cos \theta_2 dA_1 dA_2 / (r^2)$



View Factor

- 1 Energy leaving dA_1 arriving at dA_2 , $dq_{1-2} = E_{b1} \cos\theta_1 \cos\theta_2 dA_1 dA_2 / (\pi r^2)$
- 2 Energy leaving dA_2 arriving at dA_1 , $dq_{2-1} = E_{b2} \cos\theta_1 \cos\theta_2 dA_1 dA_2 / (\pi r^2)$
- 3 Net Energy transfer
 $q_{1-2} = (E_{b1} - E_{b2}) \int_{A_2} \int_{A_1} \cos\theta_1 \cos\theta_2 dA_1 dA_2 / (\pi r^2)$
- 4 The Integral is $A_1 F_{12}$ or $A_2 F_{21}$

Radiation heat transfer

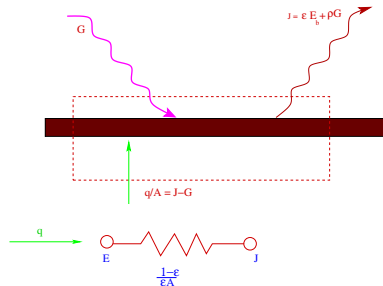


Radiation

- 1 View factor depends geometry -location

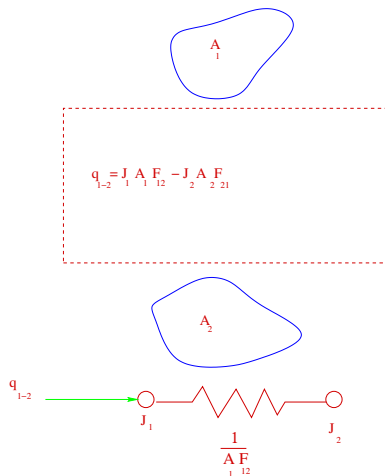
Radiation between Non Black Bodies

- 1 G = Irradiation - Total radiation incident upon a surface per unit time/unit area.
- 2 J = Radiosity - Total radiation leaving a surface per unit time/unit area.
- 3 $J = \epsilon E_b + \rho G$
- 4 $\rho = 1 - \alpha = 1 - \epsilon$
- 5 $J = \epsilon E_b + (1 - \epsilon)G$
- 6 Net energy leaving the surface $= q/A$
- 7 $q/A = J - G = \epsilon E_b + (1 - \epsilon)G - G$
- 8 $q = \frac{E_b - J}{(1 - \epsilon)/\epsilon A}$



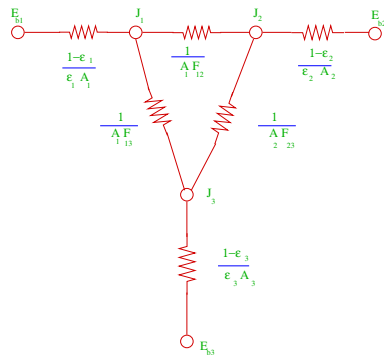
Spacial Energy Exchange

- 1 Energy leaving 1 and reaching 2
 $= J_1 A_1 F_{12}$
- 2 Energy leaving 2 and reaching 1
 $= J_2 A_2 F_{21}$
- 3 Net exchange
 $q_{1-2} = J_1 A_1 F_{12} - J_2 A_2 F_{21}$
- 4 But $A_1 F_{12} = A_2 F_{21}$
- 5 $q_{1-2} = (J_1 - J_2) A_1 F_{12} =$
 $(J_1 - J_2) A_2 F_{21}$
- 6 $q_{1-2} = \frac{J_1 - J_2}{1/A_1 F_{12}}$

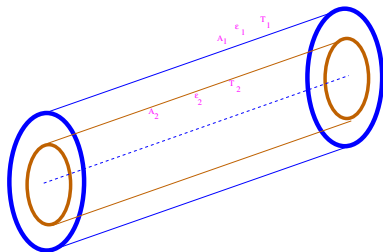
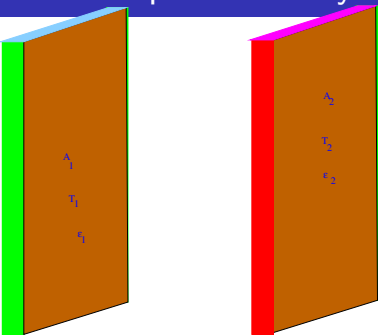


Radiation Network

- 1 Connect surface $\frac{1-\epsilon_m}{\epsilon_m A_m}$ resistance and space resistance $1/A_m F_{mn}$
- 2 Three body network
- 3 $q_{1-2} = \frac{E_{b1} - E_{b2}}{(1-\epsilon_1)/\epsilon_1 A_1 + 1/A_1 F_{12} + (1-\epsilon_2)/\epsilon_2 A_2}$
- 4 $q_{1-2} = \frac{\sigma(T_1^4 - T_2^4)}{(1-\epsilon_1)/\epsilon_1 A_1 + 1/A_1 F_{12} + (1-\epsilon_2)/\epsilon_2 A_2}$



Parallel plates and cylinders



- 1 Parallel Plates - $q_{1-2} = \frac{A\sigma(T_1^4 - T_2^4)}{(1/\epsilon_1 + 1/\epsilon_2 - 1)}$
- 2 If $\epsilon_1 = \epsilon_2$, $q_{1-2} = \frac{A\sigma(T_1^4 - T_2^4)}{(2/\epsilon - 1)}$
- 3 Concentric cylinders- $q_{1-2} = \frac{A_1\sigma(T_1^4 - T_2^4)}{1/\epsilon_1 + A_1/A_2(1/\epsilon_2 - 1)}$

Radiation shield

- 1 Shield placed between with ϵ_{31} and ϵ_{32}

- 2
$$q_{12} = \frac{A\sigma(T_1^4 - T_2^4)}{(1/\epsilon_1 + 1/\epsilon_{31} + 1/\epsilon_{32} + 1/\epsilon_2 - 2)}$$

- 3 If all the emissivity are equal

- 4
$$q_{12} = \frac{A\sigma(T_1^4 - T_2^4)}{2(2/\epsilon - 1)}$$

- 5 With N Shield

$$q_{12} = \frac{A\sigma(T_1^4 - T_2^4)}{(N+1)(2/\epsilon - 1)}$$

- 6 Ratio heat, $Q_N/Q_0 = \frac{1}{N+1}$

